

Wazuh Installation Prerequisites & System Requirements

To install WAZUH, we must follow specific prerequisites and system requirements we can find here:

<https://documentation.wazuh.com/current/quickstart.html>

This ensures that our installation goes smoothly and our system can function efficiently.

There are two specific requirements we need to follow.

1- hardware

The first is what hardware we will need to use. As you can see here:

Agents	CPU	RAM	Storage (90 days)
1-25	4 vCPU	8 GiB	50 GB
25-50	8 vCPU	8 GiB	100 GB
50-100	8 vCPU	8 GiB	200 GB

If we have between 1 to 25 agents (agents meaning endpoints like a Windows device or a Linux server, it could be anything). We need a CPU with at least 4 virtual CPUs. There must be a minimum of 8 GB of RAM and 50 GB of storage. This is if we want to analyze or collect logs for 1 to 25 agents for 90 days.

If we have 25 to 50, we will need 100 GB of storage for that, 8 GB of RAM, 8 virtual CPUs, and so on.

NB: Make sure your root partition (`/dev/mapper/ubuntu--vg-ubuntu--lv`) has sufficient headroom during node module extractions, at least 40G

```
Example:
Filesystem                Size      Used Avail Use% Mounted on
/dev/mapper/ubuntu--vg-ubuntu--lv 48G   8.5G   37G   19% /
```

If not, use this this command: `sudo lvextend -l +100%FREE /dev/mapper/ubuntu--vg-ubuntu--lv`

and this command `sudo resize2fs /dev/mapper/ubuntu--vg-ubuntu--lv`

to Expand the logical volume and resize the filesystem to claim all remaining unallocated space

2- operating system

If we talk about operating system requirements, WAZUH needs a 64-bit CPU. And It can runs on

- Amazon Linux,
- CentOS Stream 10
- Red Hat Enterprise 7,8,9,10
- Ubuntu 16.04, 18.04, 20.04, 22.04, 24.04

NB: We will use Ubuntu Server 24.04 in our lab.

NB: If you want to avoid hardware and configuration, WAZUH also offers a cloud option based on the SaaS model (<https://wazuh.com/cloud/>). So, you can also use that service.



Wazuh Installation Prerequisites & System Requirements

A step-by-step guide to prepare your system for a successful Wazuh installation

Plan | Prepare | Install | Monitor | Secure



1 Introduction

Wazuh is an open-source security platform used for threat detection, log analysis, and incident response. It works as a SIEM (Security Information and Event Management) and XDR (Extended Detection and Response) solution, helping organizations monitor and analyze security events across their IT infrastructure.



Key Benefits

- ✓ Real-time monitoring and threat detection
- ✓ Centralized log management
- ✓ Incident response and compliance
- ✓ Supports endpoints, servers, networks, and cloud

2 Official Documentation

Always refer to the official Wazuh documentation before installation:

<https://documentation.wazuh.com/current/quickstart.html>

This ensures that your installation goes smoothly and your system can function efficiently.



3 Two Main Requirements

There are two specific requirements we need to follow before installing Wazuh:

1 Hardware requirements

2 Operating system requirements



Goal

Ensure a smooth installation and optimal performance for your Wazuh environment.

4 Hardware Requirements

The hardware requirements depend on the number of agents (endpoints like Windows devices, Linux servers, etc.) and the log retention period (90 days).

Number of Agents (Endpoints)	vCPU (virtual CPUs)	RAM (Minimum)	Storage (90 days)
1 - 25	4 vCPU	8 GB	50 GB
25 - 50	8 vCPU	8 GB	100 GB
50 - 100	8 vCPU	8 GB	200 GB



Important Note

Make sure your root partition (/dev/mapper/ubuntu--vg-ubuntu--lv) has sufficient headroom during node module extractions, at least 40G.

5 Check Disk Space (Example)

Run the following command to check your disk space:

```
user@wazuh:~$ df -h
Filesystem      Size  Used Avail Use% Mounted on
/dev/mapper/ubuntu--vg-ubuntu--lv 48G  8.5G  37G  18% /
```

✓ In this example, we have 37G available, which is good.

6 Expand Disk Space (If Needed)

If you don't have enough space, use these commands to expand the logical volume and resize the filesystem:

1 Extend the logical volume:

```
sudo lvextend -l +100%FREE /dev/mapper/ubuntu--vg-ubuntu--lv
```

2 Resize the filesystem:

```
sudo resize2fs /dev/mapper/ubuntu--vg-ubuntu--lv
```

✓ These commands will claim all remaining unallocated space.

7 Operating System Requirements

Wazuh requires a 64-bit CPU and can run on the following operating systems:

- Amazon Linux
- CentOS Stream 10
- Red Hat Enterprise Linux 7, 8, 9, 10
- Ubuntu 16.04, 18.04, 20.04, 22.04, 24.04

✓ In our lab, we will use Ubuntu Server 24.04.

8 Cloud Option (SaaS)

If you want to avoid hardware and configuration, Wazuh also offers a cloud option based on the SaaS model.

<https://wazuh.com/cloud/>

You can create an account and deploy Wazuh in the cloud.

9 Lab Environment Example (Ubuntu Server 24.04)

Here is an example of a system that meets the requirements for 1 - 25 agents:

```
user@wazuh:~$ lsb_release -a
No LSB modules are available.
Distributor ID: Ubuntu
Description:    Ubuntu 24.04 LTS
Release:        24.04
Codename:       noble
```



4 vCPU



8 GB RAM



50 GB+ Storage



✓ This system is ready for Wazuh installation (1 - 25 agents).

10 Next Step

After verifying the prerequisites and system requirements, you can proceed to install Wazuh using the official guide:

<https://documentation.wazuh.com/current/quickstart.html>

Next steps will include:

- ✓ Install Wazuh server (manager)
- ✓ Install Wazuh agents (endpoints)
- ✓ Configure rules and monitoring
- ✓ Test the installation



Monitor
Detect
Respond



A Stronger Security Posture with Wazuh

Open Source | Flexible | Scalable | Trusted by Security Professionals

"Knowledge shared is security multiplied." - Tims TCHOUAMOU



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